

Volume III. Seventh Section.

Theory of the Transformation Equations.

§ 70. Transformation equations of the first stage.

The invariant equation is of great theoretical importance, partly on account of its general validity (also for even n), and partly on account of the ease with which the linear transformations can be applied to $j(\omega)$. The actual calculation of this equation, however, proves to be so complicated, and the numerical coefficients are so large, that the calculation has so far been carried through only in the simplest case $p = 2$. By contrast, by using other modular functions one can obtain far simpler transformation equations.

On the principle to be applied here we make the following remarks.

Suppose there is given some system of ν functions of ω , corresponding to the ν systems of transformation numbers a, c, ∂ , which we shall denote by

$$\Phi_{a,c,\partial} \tag{1}$$

and which are permuted among one another, isomorphically with the functions

$$j\left(\frac{c + \partial\omega}{a}\right), \tag{2}$$

by the substitutions

$$\left(\omega, -\frac{1}{\omega}\right), \quad (\omega, \omega + 1). \tag{3}$$

Then (1) is rationally expressible through (2) and through $j(\omega)$; for the function

$$F_n[x, j(\omega)] \sum_{a,c,\partial} \frac{\Phi_{a,c,\partial}}{x - j\left(\frac{c + \partial\omega}{a}\right)} = \Psi[x, j(\omega)] \tag{4}$$

remains unchanged under the substitutions (3), and is therefore, for an indeterminate x , a rational function of $j(\omega)$, and moreover an integral rational function of x , at most of degree $\nu - 1$. Here $F_n[x, j(\omega)]$ has the same meaning as in (17) of the preceding paragraph, and $\nu = \psi(n)$ is the degree of this function with respect to x . From (4), however, by putting

$$x = j\left(\frac{c + \partial\omega}{a}\right),$$

one obtains

$$\Phi_{a,c,\partial} = \frac{\Psi\left[j\left(\frac{c + \partial\omega}{a}\right), j(\omega)\right]}{F'_n\left[j\left(\frac{c + \partial\omega}{a}\right)\right]}. \tag{5}$$

Thus $\Phi_{a,c,\partial}$ belongs to the algebraic field which consists of the rational functions of $j((c + \partial\omega)/a)$ and $j(\omega)$, and we can apply the theorems of Vol. I, § 151. From these it follows that the ν quantities $\Phi_{a,c,\partial}$ are the roots of an equation of degree ν whose coefficients depend rationally on $j(\omega)$. If the ν quantities $\Phi_{a,c,\partial}$ are mutually distinct, then this equation is irreducible. Such an equation we call a transformation equation belonging to the transformation degree n and of the first stage.¹ Every other quantity of the field can be expressed rationally through such a $\Phi_{a,c,\partial}$ and through $j(\omega)$.

If the functions $\Phi_{a,c,\partial}$ have the property of remaining finite for every finite value of ω with positive imaginary part, hence for every finite $j(\omega)$, then the function $\Psi[x, j(\omega)]$ in (4) is also an

¹In the first edition I called these equations "invariant transformation equations." Klein objected to this expression (*Vorlesungen über ausgewählte Kapitel der Zahlentheorie*, autographed notebook, Göttingen 1897). I follow Klein's suggestion by now calling these equations "transformation equations of the first stage," without here going more closely into the reason for this expression.

integral function with respect to $j(\omega)$. The formula (5) could therefore fail only for those special values of ω for which two roots of the invariant equation become equal.

If the ν quantities $\Phi_{a,c,\partial}$ are not all mutually distinct, then they decompose into rows of equal numbers of quantities equal to one another, and the equation of degree ν derived from (5) is a power of an irreducible equation, which we shall occasionally also call a transformation equation (Vol. I, § 151, 2).

If the quantities $\Phi_{a,c,\partial}$ are chosen so that they do not become infinite for any finite ω with positive imaginary component, then they remain finite for every finite $j(\omega)$; from this it follows that the coefficients in the function of degree ν

$$\prod (x - \Phi_{a,c,\partial}) \quad (6)$$

are integral rational functions of $j(\omega)$, that is, the $\Phi_{a,c,\partial}$ are integral algebraic functions of $j(\omega)$.

If one arranges the functions $\Phi_{a,c,\partial}$, for instance by a suitable determination of constants still available in them, so that they also remain finite for an infinitely imaginary ω , that is, for $q = 0$, then these functions are also finite for an infinite $j(\omega)$, and the coefficients in (6) are constant. But this is possible only because the $\Phi_{a,c,\partial}$ are all equal to one and the same constant C . If $\Phi_{a,c,\partial}$ is known as a rational function of another quantity $\Psi_{a,c,\partial}$, then $\Phi_{a,c,\partial} = C$ gives either an identity or a transformation equation for $\Psi_{a,c,\partial}$.

§ 71. The transformation equations for γ_2 and γ_3 .

Transformation equations of the first stage are obtained first from consideration of the functions § 54, (4), (5):

$$\gamma_2(\omega) = \sqrt[3]{j(\omega)}, \quad \gamma_3(\omega) = \sqrt{j(\omega) - 1728}.$$

If n is not divisible by 3, then we can choose the numbers a, c, ∂ so that c is always divisible by 3.

Now, according to § 54, (15), a linear transformation

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

exerts on the function $\gamma_2(\omega)$ the influence

$$\gamma_2\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right) = e^{-\frac{2\pi i}{3}(\alpha\gamma + \beta\delta - \alpha_3 - \alpha_2^3\delta)} \gamma_2(\omega).$$

In the compositions (8), (10) of § 69, one has

$$\lambda \equiv n, \quad \delta \equiv 0 \pmod{3},$$

and then α can still be so determined that α too, and consequently c_2 , become divisible by 3.

It follows that under the two substitutions

$$(\omega, \omega + 1), \quad \left(\omega, -\frac{1}{\omega}\right)$$

the ν functions

$$\gamma_2\left(\frac{c + \partial\omega}{a}\right) \gamma_2(\omega)^{-n} \quad (1)$$

are only permuted among one another and hence are the roots of a transformation equation of the first stage.

Secondly, if n is an odd number, take c even.

For the function $\gamma_3(\omega)$, according to § 54, (15), one has

$$\gamma_3\left(\frac{\gamma + \delta\omega}{\alpha + \beta\omega}\right) = (-1)^{\alpha\gamma + \beta\delta + \gamma^3\delta} \gamma_3(\omega),$$

and in the compositions (8), (10), § 69, one has

$$\lambda \equiv 1, \quad \alpha \equiv \delta \equiv 0 \pmod{2}.$$

The ν quantities

$$\gamma_3\left(\frac{c + \partial\omega}{a}\right) \gamma_3(\omega) \quad (2)$$

therefore permute among one another and consequently likewise are the roots of a transformation equation of the first stage.

In order to form the first of these equations for the simplest case $n = 2$, one observes the relation § 54, (5):

$$\gamma_2(\omega) = \frac{f(\omega)^{24} - 16}{f(\omega)^8} = \frac{f_1(\omega)^{24} + 16}{f_1(\omega)^8} = \frac{f_2(\omega)^{24} + 16}{f_2(\omega)^8}, \quad (3)$$

from which, because

$$f_1(2\omega)f_2(\omega) = \sqrt{2} \quad [\S 34, (16)],$$

it follows that

$$\begin{aligned} \gamma_2(2\omega) &= \frac{2^8 + f_2(\omega)^{24}}{f_2(\omega)^{16}}, \\ \gamma_2\left(\frac{\omega}{2}\right) &= \frac{2^8 + f_1(\omega)^{24}}{f_1(\omega)^{16}}, \\ \gamma_2\left(\frac{\omega + 3}{2}\right) &= \frac{2^8 - f(\omega)^{24}}{f(\omega)^{16}}. \end{aligned} \quad (4)$$

Denoting these three quantities by x, x_0, x_1 , one obtains from the relations (6), (8), § 54,

$$\begin{aligned} x + x_0 + x_1 &= \gamma_2(\omega)^2, \\ xx_0 + xx_1 + x_0x_1 &= 495 \gamma_2(\omega), \\ xx_0x_1 &= -j(\omega) + 2^4 \cdot 3^3 \cdot 5^3, \end{aligned}$$

so that x, x_0, x_1 are the roots of the equation

$$x^3 - \gamma_2(\omega)^2 x^2 + 5 \cdot 9 \cdot 11 \gamma_2(\omega) x + j(\omega) - 2^4 \cdot 3^3 \cdot 5^3 = 0. \quad (5)$$

The equation whose roots are the cubes of the roots of (5) is the invariant equation for $n = 2$, and can be calculated from it without difficulty.

§ 72. Multiplier equations of the first stage.

Among the multiplier equations we shall consider here those whose roots are the different powers of P_{11} , multiplied by powers of $\gamma_2(\omega)$, $\gamma_3(\omega)$, whose coefficients depend rationally on $j(\omega)$. These equations we call multiplier equations of the first stage.²

²These equations were studied especially thoroughly by Kiepert, first in several papers in Crelle's Journal, vols. 57, 88, 95, and most fully in papers in the *Mathematische Annalen*, vols. 26, 33. Compare also F. Klein, *Mathematische Annalen*, vols. 14, 15, and the above-mentioned autographed lectures.

We consider the quantities § 68, (16):

$$P_{c,\partial,a} = i^{\frac{a-1}{2}} \left(\frac{c}{e} \right) \sqrt{n} \frac{\eta \left(\frac{c+\partial\omega}{a} \right)}{\eta(\omega)}, \quad (1)$$

and the influence exerted on them by the transformations

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

This influence is determined by the formulae (8) to (13), § 69, according to which, since $c_1 \equiv c \pmod{e}$,

$$P_{c,\partial,a} \quad \text{under } (\omega, \omega + 1) \quad \text{goes into} \quad e^{\frac{\pi i}{12}(\lambda-1)} P_{c_1,\partial,a}, \quad (2)$$

and under $(\omega, -1/\omega)$ into

$$\sqrt{\frac{\partial}{c_2}} \frac{i^{(a-a_2)/2}}{\sqrt{-i\omega}} \left(\frac{c}{e} \right) \left(\frac{c_2}{e_2} \right) E(\alpha, \beta, c_2 + \hat{c}_2 \omega \text{ over } a_2) P_{c_2,\partial_2,a_2},$$

where E has the meaning assigned in § 38, (15). But by § 69, (11), (12),

$$\sqrt{\partial} \sqrt{\alpha + \beta \frac{c_2 + \hat{c}_2 \omega}{a_2}} = \sqrt{\partial_2 \omega},$$

and therefore, if we put

$$E \left(\alpha, \beta, \frac{c_2 + \hat{c}_2 \omega}{a_2} \right) = r \sqrt{-i\omega} \sqrt{\frac{c_2}{\partial}},$$

then

$$r = \left(\frac{\alpha}{\beta} \right) i^{\frac{1-\beta}{2}} e^{\frac{\pi i}{12} [\partial(\alpha+\partial) - (\partial^2-1)\alpha\gamma]}$$

is a twenty-fourth root of unity, and under the interchange

$$\left(\omega, -\frac{1}{\omega} \right)$$

$P_{c,\partial,a}$ goes into

$$r i^{(a-a_2)/2} \left(\frac{c}{e} \right) \left(\frac{c_2}{e_2} \right) P_{c_2,\partial_2,a_2}.$$

From this follows, as above, the conclusion:

1. For every arbitrary n , the quantities

$$P_{c,\partial,a}^{24}$$

are the roots of a transformation equation.

If n is not divisible by 3, then c, c_2, c_1 can be assumed divisible by 3. Then

$$\lambda \equiv n, \quad \alpha \equiv \partial \equiv 0 \pmod{3}.$$

Thus, as follows from § 38, (15), r is an eighth root of unity; if one further observes

$$\gamma_2(\omega + 1) = e^{-2\pi i/3} \gamma_2(\omega), \quad \gamma_2 \left(-\frac{1}{\omega} \right) = \gamma_2(\omega), \quad [\S 54, (14)],$$

then we have the theorem:

2. If n is not divisible by 3 and c is divisible by 3, then the quantities

$$P_{c,\partial,a}^8 \gamma_2(\omega)^{n-1}$$

are roots of a transformation equation.

To make the application to the simplest case $n = 2$, we put

$$x = 16 \left(\frac{\eta(2\omega)}{\eta(\omega)} \right)^8, \quad x_0 = \frac{\eta\left(\frac{\omega}{2}\right)^8}{\eta(\omega)^8}, \quad x_1 = \frac{\eta\left(\frac{\omega+3}{2}\right)^8}{\eta(\omega)^8}, \quad (3)$$

and then x, x_0, x_1 are nothing other than our functions

$$f_2(\omega)^8, \quad f_1(\omega)^8, \quad -f(\omega)^8. \quad [\S 34, (9)]$$

These, as we have already seen earlier, § 54, (8), are the roots of the cubic equation

$$x^3 - \gamma_2(\omega)x + 16 = 0. \quad (4)$$

The circumstance that $f_2^8, f_1^8, -f^8$ themselves are roots of a transformation equation for transformation degree 2 explains the phenomenon that upon adjoining these quantities, hence also upon adjoining α_2 , the transformation equations belonging to even n become reducible.

If n is an odd number, take c divisible by 4; then by § 69, (9), (11), (12),

$$\lambda \equiv n, \quad \alpha \equiv 0, \quad \partial \equiv 0, \quad \beta \equiv a\partial_2, \quad \gamma \equiv -a_2\alpha \pmod{4},$$

and therefore

$$r^6 = (-1)^{(\beta-1)/2} = (-1)^{(n-1)/2}(-1)^{(a-a_2)/2}.$$

With regard to

$$\gamma_3(\omega + 1) = -\gamma_3(\omega), \quad \gamma_3\left(-\frac{1}{\omega}\right) = -\gamma_3(\omega), \quad [\S 54, (14)],$$

it follows that

3. If n is odd and c divisible by 4, then the quantities

$$P_{c,\partial,a}^6 \gamma_3(\omega)^{(n-1)/2}$$

are roots of a multiplier equation. If $n \equiv 1 \pmod{4}$, the same holds for $P_{c,\partial,a}^6$.

As example choose $n = 3$ and put

$$x = 27 \left(\frac{\eta(3\omega)}{\eta(\omega)} \right)^6, \quad x_0 = - \left(\frac{\eta\left(\frac{\omega}{3}\right)}{\eta(\omega)} \right)^6, \quad x_1 = - \left(\frac{\eta\left(\frac{4+\omega}{3}\right)}{\eta(\omega)} \right)^6, \quad x_2 = - \left(\frac{\eta\left(\frac{8+\omega}{3}\right)}{\eta(\omega)} \right)^6. \quad (5)$$

The coefficients of the equation which has these quantities as roots are rational functions of $\gamma_3(\omega)$; and since none of the quantities (5) becomes infinite for a finite value of γ_3 , they are integral functions of γ_3 . According to 3. we can therefore set this equation in the form

$$x^4 + A_1\gamma_3x^3 + A_2x^2 + A_3\gamma_3x + A_4 = 0,$$

where A_1, A_2, A_3, A_4 are integral rational functions of $j(\omega)$. First one obtains A_4 as the product $xx_0x_1x_2$, which cannot vanish for any value of $j(\omega)$ and therefore must be constant. From the initial terms of the expansion § 24,

$$x = 27q \cdots, \quad x_0 = -q^{-1/3} \cdots, \quad x_1 = -e^{2\pi i/3} q^{-1/3} \cdots, \quad x_2 = -e^{-2\pi i/3} q^{-1/3} \cdots, \quad \gamma_3 = -q^{-1} \cdots, \quad (6)$$

one therefore finds

$$A_4 = -27.$$

Furthermore,

$$-A_1\gamma_3 = x + x_0 + x_1 + x_2.$$

Since the right hand side is not even infinite of the first order for an infinite γ_3 , that is, for $q = 0$, A_1 must vanish.

From

$$-A_3\gamma_3 = xx_0x_1x_2 \left(\frac{1}{x} + \frac{1}{x_0} + \frac{1}{x_1} + \frac{1}{x_2} \right)$$

one obtains from (6), for A_3 , the constant value 1. In order, however, to determine A_2 , we must go one term further in the expansion. We put in

$$x_0^4 + \gamma_3x_0 - 27 = -A_2x_0^2$$

for x_0 the expansion

$$x_0 = -q^{-1/3} + 6q^{1/3} + \dots,$$

and find $A_2 = 18$, so that the sought equation is

$$x^4 + 18x^2 + \gamma_3x - 27 = 0. \quad (7)$$

If n is odd and not divisible by 3, take c divisible by 12. Then

$$\lambda \equiv n, \quad \alpha \equiv 0, \quad \partial \equiv 0, \quad \beta \equiv a\partial_2, \quad \gamma \equiv -a_2\alpha \pmod{12},$$

and one obtains

$$r^2 = (-1)^{(\beta-1)/2} = (-1)^{(n-1)/2}(-1)^{(a-a_2)/2},$$

hence the theorem:

4. If n is relatively prime to 6 and c divisible by 12, then the quantities

$$P_{c,\partial,a}^2 \gamma_2(\omega)^{n-1} \gamma_3(\omega)^{(n-1)/2}$$

are roots of a multiplier equation of the first stage.

Since the functions P^2 become neither infinite nor zero for any finite value of $j(\omega)$, we conclude that the coefficients of the equation whose roots are the P^2 are integral rational functions of γ_2, γ_3 , and that the last coefficient is a constant.

If n is a prime number p , then the roots of this equation are

$$x = p \left(\frac{\eta(p\omega)}{\eta(\omega)} \right)^2, \quad x_h = (-1)^{(p-1)/2} e^{2\pi i h/p} \left(\frac{\eta\left(\frac{12h+\omega}{p}\right)}{\eta(\omega)} \right)^2,$$

where $h = 0, 1, \dots, p-1$, and the initial terms of the expansions are the following:

$$x = pq^{(p-1)/6} \dots, \quad x_h = (-1)^{(p-1)/2} e^{2\pi i h/p} q^{-(p-1)/6p} \dots,$$

from which the value $(-1)^{(p-1)/2}p$ is obtained for the last coefficient. From the fact that the first term in the expansion of $j(\omega)$ by powers of q has coefficient 1, one readily concludes that the numerical coefficients in these equations are rational integers.

For $p = 5$ the equation in question has the form

$$x^6 + A_1\gamma_2^2x^5 + A_2\gamma_2x^4 + A_3x^3 + A_4\gamma_2^2x + 5 = 0,$$

where A_1, A_2, A_3, A_4 are integral rational functions of $j(\omega)$ and, as is easy to see, constants.

From the initial terms one obtains immediately

$$A_1 = 0, \quad A_2 = 0, \quad A_4 = -1.$$

But in order to determine A_3 , one goes in the expansion of x_0 to the next following term:

$$x_0 = q^{-7/15}(1 - 2q^{2/5} + \dots),$$

from which $A_3 = 10$ is obtained; thus, for $n = 5$, the multiplier equation is

$$x^6 + 10x^3 - \gamma_2 x + 5 = 0. \quad (8)$$

In the same way the equations for $p = 7$, $p = 11$ are calculated, by using the expansions of x_0 :

$$p = 7: \quad x_0 = -q^{-1/7}(1 - 2q^{2/7} - q^{4/7} + 2q^{6/7} \dots),$$

$$p = 11: \quad x_0 = -q^{-5/33}(1 - 2q^{4/11} - q^{6/11} + 2q^{8/11} + q^{10/11} + 2q^{12/11} + \dots),$$

while from $\gamma_2(\omega), \gamma_3(\omega)$ only the first terms are ever needed. Thus one finds

$$p = 7: \quad x^8 + 7 \cdot 2x^6 + 7 \cdot 9x^4 + 7 \cdot 10x^2 + \gamma_3 x - 7 = 0, \quad (9)$$

$$p = 11: \quad x^{12} - 11 \cdot 90x^8 + 11 \cdot 40\gamma_2 x^6 + 11 \cdot 15\gamma_3 x^5 + 11 \cdot 2\gamma_2^2 x^2 + \gamma_2 \gamma_3 x - 11 = 0. \quad (10)$$

If n is an odd square, take c divisible by 8. Then $a : e$ and $\partial : e$ are likewise squares. Further,

$$\alpha \equiv \partial \equiv 0 \pmod{8}, \quad \lambda \equiv 1 \pmod{8}, \quad (11)$$

$$a = \beta \partial_2, \quad a_2 \equiv \beta \partial, \quad a \equiv \partial \equiv e \equiv 1, \quad c = \partial \hat{\partial}_2, \quad c_2 = -\alpha \partial, \quad a_2 \equiv \partial_2 \equiv e_2 \equiv 1 \pmod{8}. \quad (12)$$

The root of unity r in (2) receives the value

$$r = \left(\frac{\alpha}{\beta}\right) i^{(1-\beta)/2} e^{\frac{2\pi i}{3}(\alpha+\partial) - (\partial-1)\alpha\gamma/8}. \quad (13)$$

But by (12) β is divisible both by e and by e_2 , and $\beta e e_2$ is a square; hence

$$\left(\frac{\alpha}{\beta}\right) = \left(\frac{\alpha}{e e_2}\right) = \left(\frac{\alpha}{e}\right) \left(\frac{\alpha}{e_2}\right) = \left(\frac{\partial}{e}\right) \left(\frac{\alpha}{e_2}\right) \quad (\text{because } \alpha \partial \equiv 1 \pmod{e}).$$

Further by (12)

$$\left(\frac{\partial}{e}\right) = \left(\frac{c}{e}\right) \left(\frac{\partial_2}{e}\right), \quad \left(\frac{\alpha}{e_2}\right) = \left(\frac{c_2}{e_2}\right) \left(\frac{-\partial}{e_2}\right),$$

and

$$\left(\frac{c_2}{e}\right) = \left(\frac{c_2}{e_2}\right), \quad \left(\frac{-\partial}{e_2}\right) = \left(\frac{-e}{e_2}\right), \quad \left(\frac{e_2}{e}\right) \left(\frac{-e}{e_2}\right) = (-1)^{(e+1)(e_2-1)/4},$$

therefore

$$\left(\frac{\alpha}{\beta}\right) = (-1)^{(e+1)(e_2-1)/4} \left(\frac{c}{e}\right) \left(\frac{c_2}{e_2}\right),$$

and

$$i^{(1-\beta)/2} = i^{(1-e)/2}, \quad i^{(a-a_2)/2} = i^{(e-e_2)/2}.$$

By the interchanges (2), $P_{c,\partial,a}$ therefore goes into

$$P_{c,\partial,a} \quad \text{under } (\omega, \omega+1) \text{ into } e^{\frac{2\pi i}{8} \frac{\lambda-1}{2}} P_{c_1,\partial,a}, \quad \text{and under } \left(\omega, -\frac{1}{\omega}\right) \text{ into } e^{\frac{2\pi i}{8} \frac{\{\beta(\alpha+\partial) - (\partial^2-1)\alpha\gamma\}}{4}} P_{c_2,\partial_2,a_2}. \quad (14)$$

If n is still not divisible by 3, take c divisible by 3, whereby

$$\lambda \equiv 1, \quad \alpha \equiv \partial \equiv 0 \pmod{3},$$

and the third roots of unity occurring in (14) receive the value 1.

From this the following theorems result:

5. If n is an odd square and c divisible by 8, then the quantities

$$P_{c,\partial,a}^3$$

and, if n is an odd square not divisible by 3 and c is divisible by 24, then the quantities

$$P_{c,\partial,a}$$

are roots of multiplier equations of the first stage.

The first examples are $n = 9$, $n = 25$.

For the formation of these equations one can proceed in different ways. We shall here, following Kiepert,³ take the path of expressing, by § 69, the roots of a transformation equation belonging to degree p rationally through those for degree p^2 , and then inserting this expression into the transformation equation belonging to p .

According to 3., formula (7), the equation

$$x^4 + 18x^2 + \gamma_3(\omega)x - 27 = 0 \quad (15)$$

is satisfied by the two functions

$$-\left(\frac{\eta\left(\frac{\omega}{3}\right)}{\eta(\omega)}\right)^6, \quad 27\left(\frac{\eta(3\omega)}{\eta(\omega)}\right)^6. \quad (16)$$

If we denote the first of these values by x , then the second passes, under the substitution $(\omega, \omega/3)$, into $-27/x$, and therefore x also satisfies the equation

$$27^3 + 18 \cdot 27x^2 - \gamma_3\left(\frac{\omega}{3}\right)x^3 - x^4 = 0. \quad (17)$$

Now put

$$y = \left(\frac{\eta\left(\frac{\omega}{9}\right)}{\eta(\omega)}\right)^3. \quad (18)$$

Then x passes under the substitution $(\omega, \omega/3)$ into y^2/x , so that one also obtains the equation

$$y^8 + 18y^4x^2 + \gamma_3\left(\frac{\omega}{3}\right)y^2x^3 - 27x^4 = 0, \quad (19)$$

and by eliminating $\gamma_3(\omega/3)$ from (17) and (19), after removing the factor $y^2 - 27$, one obtains

$$x^4 - 18x^2y^2 - y^2(y^4 - 27y^2 + 27^2) = 0. \quad (20)$$

Solving this quadratic equation for x^2 , one obtains

$$x^2 = y^3 + 9y^2 + 27y = (y + 3)^3 - 27, \quad (21)$$

where the sign is decided by inserting the initial terms of the expansions, most simply since this equation is also satisfied, according to § 69, by

$$x = 27\left(\frac{\eta(3\omega)}{\eta(\omega)}\right)^6, \quad y = 27\left(\frac{\eta(9\omega)}{\eta(\omega)}\right)^3,$$

putting

$$x = 27q + \dots, \quad y = 27q^2 + \dots.$$

If in (15) the first power of x is isolated and the equation is squared, then insertion of (21) gives the sought multiplier equation for the ninth transformation degree. It receives a simpler form if one puts

$$t = y + 9, \quad (22)$$

³Zur Transformationstheorie der elliptischen Funktionen, Crelle's Journal, vols. 87, 88, 95.

namely

$$[j(\omega) - 27 \cdot 64](t - 27) = (t^2 - 36t + 27 \cdot 8)^2, \quad (23)$$

or

$$j(\omega)(t - 27) = t(t - 24)^3. \quad (24)$$

Quite similarly one can proceed for the twenty-fifth transformation degree.

If we put

$$x = \left(\frac{\eta(\frac{\omega}{5})}{\eta(\omega)} \right)^2, \quad y = \left(\frac{\eta(\frac{\omega}{25})}{\eta(\omega)} \right), \quad (25)$$

then first, by 4., (8),

$$x^6 + 10x^3 - \gamma_2(\omega)x + 5 = 0, \quad (26)$$

from which, as above, the two equations

$$5^5 + 10 \cdot 5^2 x^3 - \gamma_2\left(\frac{\omega}{5}\right) x^5 + x^6 = 0,$$

$$y^{11} + 10y^6 x^3 - \gamma_2\left(\frac{\omega}{5}\right) y^2 x^5 + 5x^6 = 0,$$

and by eliminating $\gamma_2(\omega/5)$,

$$x^6 - 10x^3 y^2(5 + y^2) - y^2(y^8 + 5y^6 + 5^2 y^4 + 5^3 y^2 + 5^4) = 0.$$

Solving this equation for x^3 gives

$$x^3 = y^5 + 5y^4 + 15y^3 + 25y^2 + 25, \quad (27)$$

and if, for abbreviation, we denote the right hand side of this equation by $\chi(y)$, then from (26)

$$\gamma_2(\omega)y^2 = \chi(y)^2 + 10\chi(y) + 5^2. \quad (28)$$

This is the sought equation of degree 30 for y .